Customer Shopping Behavior Analysis

1. Project Overview

This project analyzes customer shopping behavior using transactional data to understand purchasing patterns, customer preferences, and decision-making factors.

The insights derived help businesses enhance marketing strategies, improve customer experience, and optimize overall performance.

2. Dataset Summary

Total Records: 3,900

Total Features: 18

Key Data Includes:

- Customer Details: Age, Gender, Location, Subscription Status

- Purchase Details: Product, Category, Purchase Amount, Season, Size, Color

- Behavioral Data: Discounts, Promo Codes, Previous Purchases, Purchase Frequency, Ratings, Shipping Type

Data Quality Note: Missing values were identified in the Review Rating column.

3. Data Preparation & Analysis (Python)

Data was cleaned and prepared using Python to ensure accuracy and consistency.

- Data Loading: Dataset imported using pandas

- Initial Exploration: Structure checked using info() and describe()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 19 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   customer_id             3900 non-null   int64
 1   age                     3900 non-null   int64
 2   gender                  3900 non-null   object
 3   item_purchased          3900 non-null   object
 4   category                3900 non-null   object
 5   purchased_amount        3900 non-null   int64
 6   location                3900 non-null   object
 7   size                    3900 non-null   object
 8   color                   3900 non-null   object
 9   season                  3900 non-null   object
 10  review_rating           3900 non-null   float64
 11  subscription_status     3900 non-null   object
 12  shipping_type           3900 non-null   object
 13  discount_applied        3900 non-null   object
 14  previous_purchases      3900 non-null   int64
 15  payment_method          3900 non-null   object
 16  frequency_of_purchases  3900 non-null   object
 17  age_group               3900 non-null   category
 18  purchase_frequency_days 3900 non-null   int64
dtypes: category(1), float64(1), int64(5), object(12)
memory usage: 552.6+ KB
```

	Customer ID	Age	Purchase Amount (USD)	Review Rating	Previous Purchases
count	3900.000000	3900.000000	3900.000000	3863.000000	3900.000000
mean	1950.500000	44.068462	59.764359	3.750065	25.351538
std	1125.977353	15.207589	23.685392	0.716983	14.447125
min	1.000000	18.000000	20.000000	2.500000	1.000000
25%	975.750000	31.000000	39.000000	3.100000	13.000000
50%	1950.500000	44.000000	60.000000	3.800000	25.000000
75%	2925.250000	57.000000	81.000000	4.400000	38.000000
max	3900.000000	70.000000	100.000000	5.000000	50.000000

- Missing Values: Filled using median values grouped by product category

- Column Standardization: Converted to snake_case format

- Feature Engineering:

- age_group for customer segmentation
- purchase_frequency_days for behavioral analysis

- Data Validation: Removed duplicates and redundant columns

- Database Integration: Cleaned data stored in PostgreSQL for further analysis

4. Data Analysis (SQL)

SQL queries were used to extract key business insights:

4.1 Gender-wise Revenue Insights – Analyzed and contrasted the overall revenue generated by male and female customer segments to understand their contribution to total sales.

	gender text	sum numeric
1	Female	75191
2	Male	157890

4.2 Discount Users with High Spending Behavior – Determined which customers, despite using discounts, maintained spending levels higher than the average purchase value.

	customer_id bigint	purchased_amount bigint
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62
11	24	88

Total rows: 839 Query complete 00:00

4.3 Top 5 Highest Rated Products – Found the five products that have the highest average customer ratings.

	item_purchased text	Average product rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

4.4 Shipping Type Comparison – Compared the average amount spent by customers using Standard shipping and Express shipping.

	shipping_type text	round numeric
1	Standard	58.46
2	Express	60.48

4.5 Subscribers vs Non-Subscribers – Compared how much subscribers and non-subscribers spend on average and their total revenue contribution.

	subscription_status text	Total customers bigint	avg_spend numeric	total_revenue numeric
1	No	2847	59.87	170436
2	Yes	1053	59.49	62645

4.6 Discount-Dependent Products – Found the top 5 products that are mostly bought using discounts.

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.00
3	Coat	49.00
4	Sweater	48.00
5	Pants	47.00

4.7 Customer Segmentation – Divided customers into New, Returning, and Loyal groups based on their purchase history.

	customer_segment text	total_customers bigint
1	Loyal	3116
2	New	83
3	Returning	701

4.8 Top 3 Products per Category – Identified the three most purchased products in each category.

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessori...	Jewelry	171
2	2	Accessori...	Sunglasses	161
3	3	Accessori...	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Socks	159
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Boots	144
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

4.9 Repeat Buyers and Subscriptions – Checked if customers who made more than 5 purchases are more likely to subscribe.

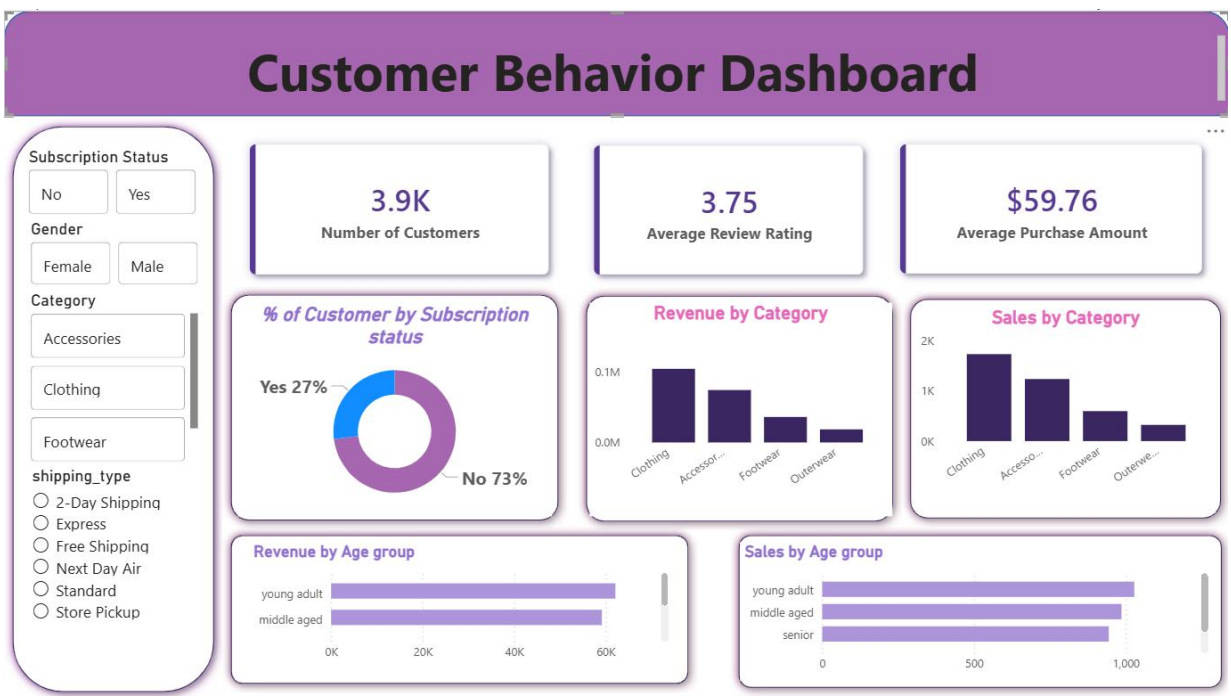
	subscription_status text	total_repeat_buyers bigint
1	No	2518
2	Yes	958

4.10 Revenue by Age Group – Calculated how much revenue is generated by each age group.

	age_group text	total_customers bigint	total_revenue numeric
1	young adult	1028	62143
2	senior	944	55763
3	adult	942	55978
4	middle aged	986	59197

5. Dashboard (Power BI)

An interactive dashboard was developed to visualize insights effectively.



It enables quick understanding of trends, comparisons across segments, and data-driven decision-making.

6. Business Recommendations

- Enhance subscription programs with exclusive benefits

- Develop loyalty programs to retain customers

- Optimize discount strategies to balance profit and sales

- Promote high-performing and highly rated products

- Focus marketing efforts on high-value customer segments

Conclusion

This project demonstrates the transformation of raw data into actionable insights using:

- Python for data cleaning and preparation

- SQL for analytical processing

- Power BI for visualization and reporting